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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-13	Team No-13	Project Title- Mini Search Engine/ Information-Retrieval System

1. Abstract

- The Mini Search Engine / Information-Retrieval System is a document retrieval application developed to efficiently search and retrieve information from a collection of documents. The primary objective of this project is to provide fast, accurate, and intelligent search results using fundamental Information Retrieval (IR) and Data Structures and Algorithms (DSA) techniques. Instead of scanning every document for each user query, the system first builds an inverted index that stores the relationship between terms and the documents in which they appear, enabling efficient keyword lookup.
- The system supports four major types of queries: exact keyword search, phrase or substring search, fuzzy search, and document similarity search. Exact keyword search retrieves documents containing the specified word, while phrase search identifies documents containing an exact sequence of words. Fuzzy search handles spelling mistakes by suggesting the closest matching keyword using an edit-distance algorithm. Document similarity search compares the content of documents using TF-IDF and cosine similarity to recommend documents with similar topics or content.
- All core algorithms are implemented manually without relying on external search-engine libraries, allowing a clear understanding of indexing, searching, ranking, and similarity computation. The project also evaluates the time and space complexity of each algorithm and discusses the trade-offs involved in designing scalable search systems.
- This project demonstrates the practical implementation of Information Retrieval concepts, including text preprocessing, indexing, query processing, ranking, and similarity analysis. It provides a simple yet effective search engine capable of handling multiple search scenarios and serves as a learning model for understanding how modern search engines organize, retrieve, and rank information efficiently from large document collections.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520080045	U. Sindhuja	
2520080024	P. Bhagya Sri	
2520090051	Santosh	

3. Frameworks / Platforms / Software Used

Category	**Software / Tool Used**
Programming Language	Java (JDK 17 or above)
IDE / Development Platform	Visual Studio Code / IntelliJ IDEA / Eclipse
User Interface	Java Swing (GUI) or Java Console Application
Data Structures	HashMap, ArrayList, LinkedList (implemented where required)
File Storage	Text Files (.txt) for document corpus
Algorithms Implemented	Inverted Index, String Matching, Edit Distance, TF-IDF, Cosine Similarity
Operating System	Windows 10/11
Version Control (Optional)	Git and GitHub

Guide Name and Signature (remarks if any) _____